

Restrained apparatus

Work: DELIGNE_D034_CYCLES_HODGE_ABSOLUS

Returned session: S02

Canonical normalization: 2026-08-24.

The authority is the 12-page NUMDAM/SMF PDF. Physical page 1 is copy matter; physical pages 2–12 are the article, printed pages 23–33. The comparator and all inherited candidate/diagram material remain locator-only and ZERO_ACCEPTED.

A001 — COPY_MATTER (physical 1).

Physical page 1 is the native NUMDAM/SMF cover. It gives Pierre Deligne, the article title, the parenthetical credit Rédigé par J. L. Brylinski, and the citation Mémoires de la S. M. F., 2e série, tome 2 (1980), p. 23–33. It identifies the NUMDAM item at http://www.numdam.org/item?id=MSMF_1980_2_2_23_0, states copyright Mémoires de la S. M. F., 1980, tous droits réservés, summarizes archive-access and reuse conditions, and records digitization under the Numérisation de documents anciens mathématiques programme.

Decision. The entire cover is recorded as copy matter and excluded from both article readers; no access, copyright, or digitization prose is imported into the article body.

A002 — PUBLICATION (physical 2 / printed 23).

The article opens with SOCIÉTÉ MATHÉMATIQUE DE FRANCE; 2e série. Mémoire n° 2, 1980, p. 23–33.

Decision. Replayed as printed; the inherited Astérisque identification is rejected.

A002A — PAGE_FURNITURE (physical 2 / printed 23).

The article scans carry alternating running heads and the printed folios 23 through 33.

Decision. The edition preserves every source page boundary and records the physical-to-printed mapping; running heads and folios are treated as page furniture rather than repeated in the reader bodies.

A003 — ROLE (physical 2 / printed 23).

Pierre Deligne is the author; the exact source credit is rédigé par J. L. Brylinski.

Decision. Brylinski remains source redactor, not coauthor; English uses “written up by”.

A004 — NOTATION (physical 3 / printed 24).

The Betti Hodge decomposition uses H^B and the p -comma- q Hodge summands. The analytic exact sequence carries the integral, Lie, and exp labels.

Decision. Symbols were checked against the authority; no modernization was made beyond legible typesetting.

A005 — NOTATION (physical 4 / printed 25).

The printed Betti/de Rham comparison tensors both realizations with C and gives no tensor-over- Q subscript.

Decision. No scalar-field subscript was inserted; the candidate normalization is rejected.

A006 — AMBIGUITY (physical 4 / printed 25).

The projection sentence reads gamma tensor x and subsequently x minus m .

Decision. Both readings are retained literally; no silent mathematical emendation was made.

A006A — SOURCE_READING (physical 4 / printed 25).

The Hodge-filtration display prints the strict index condition p -prime greater than p .

Decision. The strict sign is retained in both editions; the conventional weak inequality is not substituted.

A007 — NOTATION (physical 5 / printed 26).

The Tate-motive comparisons and the exterior-square comparison omit scalar subscripts. The Legendre relation prints explicit multiplication points between omega and eta.

Decision. Replayed symbol for symbol, including the multiplication points.

A008 — SOURCE_READING (physical 6 / printed 27).

Example 3 describes both period matrices as n by $2n$, despite the adjacent g -indexing and a basis indexed through $2g$. Its alternating-form and matrix identities also print explicit multiplication points.

Decision. The source reading n by $2n$ is retained without conjectural emendation; pairing commas, multiplication points, and the source order $2i$ πi are preserved.

A008A — WORDING (physical 6 / printed 27).

The source says S is the set of ces plongements complexes, not ses plongements complexes.

Decision. The diplomatic French retains ces; the English renders it literally as these complex embeddings rather than silently correcting the source.

A009 — LAYOUT (physical 7 / printed 28).

The complete H -phi definition includes the C -factor indexed by sigma and both components of its Hodge type.

Decision. The display was rebroken over legible lines; no clause is clipped.

A010 — NOTATION (physical 7 / printed 28).

On the sigma component of the de Rham realization, x in E acts through the inverse embedding sigma-inverse of x , not sigma of x .

Decision. The inverse-embedding reading is retained; the inherited candidate disagreement is logged.

A011 — TABLE (physical 8 / printed 29).

Printed page 29 contains a two-column period/motive table with four numbered identities and four motive relations.

Decision. Recreated as a native two-column table; all entries checked.

A012 — SOURCE_READING (physical 8 / printed 29).

After the table, the source uses the subscript K on p , although the preceding formula uses the subscript E .

Decision. The subscript K is retained without emendation; the apparent notation shift is recorded rather than corrected.

A013 — NOTATION (physical 9 / printed 30).

The absolute-Hodge-cycle definition uses X with ell-adic coefficients and a Tate twist. It prints neither an etale subscript nor a base change to X over the algebraic closure.

Decision. French preserves the printed X notation, and English follows it; no base-change notation is inserted.

A014 — LABEL (physical 10 / printed 31).

The source prints unnumbered Corollaire (Borovoi).

Decision. No automatic corollary number is generated.

A015 — NOTATION (physical 10 / printed 31).

The Hodge group is the Zariski closure of the image of the multiplicative group under μ , and that same image occurs in the rational-function criterion.

Decision. The multiplicative-group notation is retained; it is not normalized to the nonzero complex numbers.

A016 — PROOF_ENDING (physical 11 / printed 32).

The first proof ends with the explicit C.Q.F.D.; the second proof has no printed terminal symbol.

Decision. No automatic proof squares are added; English translates the explicit ending as Q.E.D.

A017 — INEQUALITY (physical 11 / printed 32).

The source gives strict upper and lower inequalities: dimension of G is less than g plus 1, and greater than 2 plus the base-2 logarithm of g .

Decision. The strict signs are retained; the candidate weak-inequality normalizations are rejected.

A018 — TERMINAL (physical 12 / printed 33).

Printed page 33 contains References [1]-[7], both addresses, a large blank remainder, folio 33, and physical EOF.

Decision. References and addresses are included; blank remainder and EOF are recorded but not filled with editorial matter.

Final audit. The full 12-physical-page mapping, cover exclusion, all 11 article page units, formulae, the page-29 table, labels, references, addresses, blank remainder, and physical EOF were rechecked in a clean nonpatching audit. All inherited branches remain ZERO_ACCEPTED.